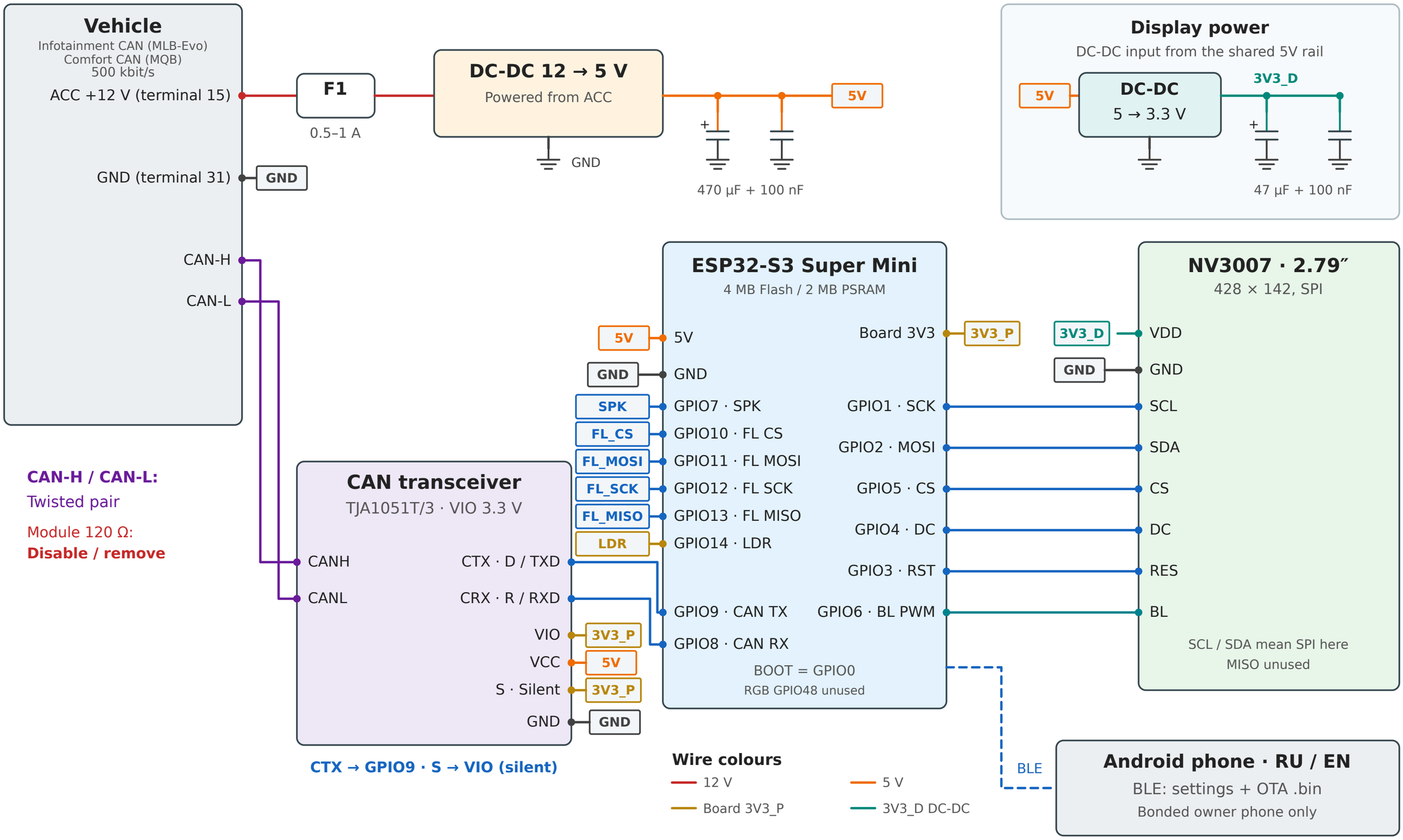
HUD — ESP32-S3 Super Mini + NV3007 2.79″

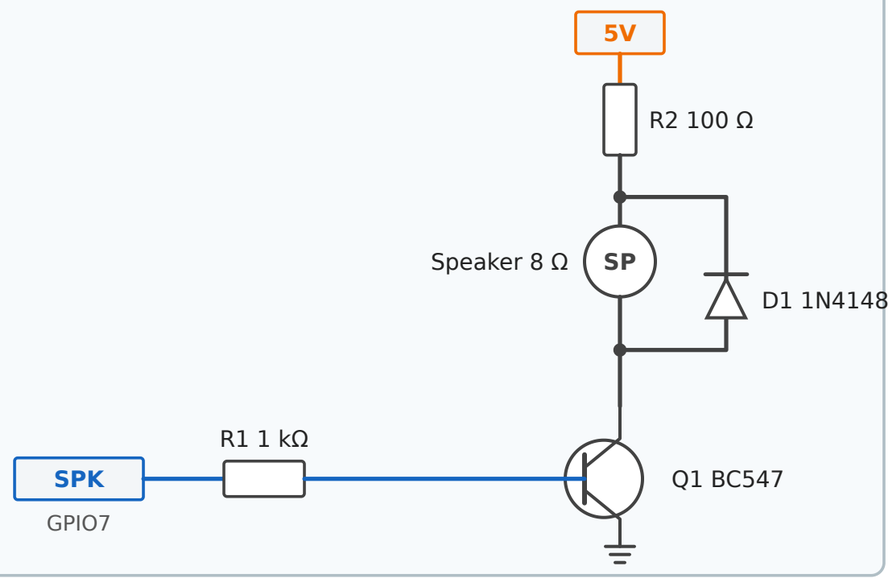
428 × 142 · Firmware GPIO mapping · Display via 5 → 3.3 V DC-DC · TJA1051T/3 CAN module

Based on gateway_supermini.svg · Receive-only CAN · Phone BLE settings and OTA



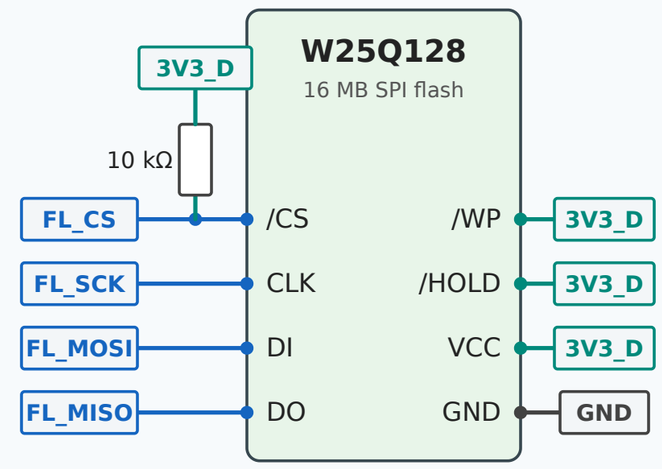
Speaker

Passive 8 Ω speaker through an NPN transistor



W25Q128 flash

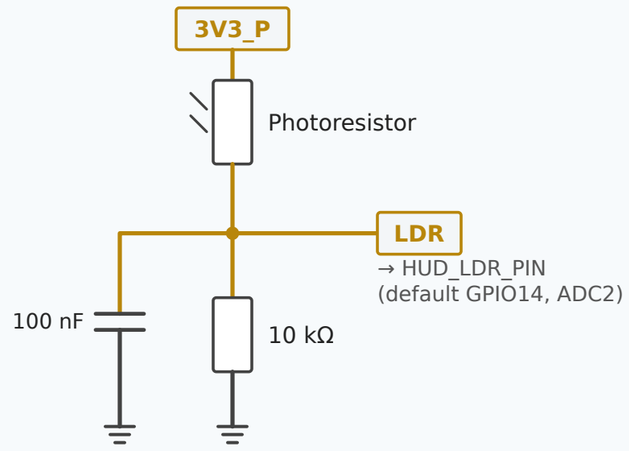
Camera database · powered from 3V3_D like the display



FL_CS = GPIO10 · FL_MOSI = GPIO11
FL_SCK = GPIO12 · FL_MISO = GPIO13

Light sensor (optional)

Used when the car gives no RLS_01 brightness



Pin is chosen by HUD_LDR_PIN in hud_config.h
ADC2 is fine: the firmware does not use Wi-Fi

Connections and notes

All GNDs common. LCD VDD → 3V3_D from 5 → 3.3 V DC-DC; input → 5V. CAN power from ESP32-S3: VCC → 5V; VIO → board 3V3.
BL / GPIO6 is a logic-level backlight control. If BL directly powers LEDs, use an external driver.
CAN: CRX → GPIO8, CTX → GPIO9 (TWAI TX). Receive-only: TWAI LISTEN_ONLY + S → VIO (silent).
Capacitors: 100 µF + 100 nF at board 3V3; 100 nF at CAN VCC; 47 µF + 100 nF at LCD VDD.
Use TJA1051T/3 with VIO: S → VIO (Silent). A version without /3 and with 5 V logic cannot be connected directly.
BOOT (GPIO0): hold for 5 seconds, then release. Pins are shown logically, not in their physical board positions.
Speaker: GPIO7 → 1 kΩ → BC547 base; speaker + 100 Ω from 5V to the collector; 1N4148 across the speaker (cathode → 5V).
W25Q128 on its own SPI bus: GPIO10 CS, GPIO11 MOSI, GPIO12 SCK, GPIO13 MISO. Power from 3V3_D like the display; CS pull-up 10 kΩ to 3V3_D; /WP and /HOLD → 3V3_D; 100 nF at VCC.
Light sensor (optional): photoresistor 3V3_P → pin, 10 kΩ pin → GND, 100 nF pin → GND. Pin set by HUD_LDR_PIN (default GPIO14, ADC2).